RAJIV GANDHI UNIVERSITY OF KNOWLEDGE AND TECHNOLOGIES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SUMMER INTERNSHIP REPORT

AI-POWERED SALOON APP

Submitted in partial fulfilment of the Summer Internship Programme

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Department	Computer Science and Engineering
Internship Organization	Saundarya App, Mumbai
Internship Type	Paid Internship
Domain	Flutter App Development
Duration	1 May 2026 to 30 June 2026
Project Internal Guide	Mr. A. Mahendra

Submitted by: Shaik Hasmi roohy

Academic Year: 2025-2026

DECLARATION

I, **Shaik Hasmi roohy**, bearing ID No. **R210448**, student of the Department of Computer Science and Engineering, Rajiv Gandhi University of Knowledge and Technologies, hereby declare that this summer internship report titled **"AI-Powered Saloon App"** is a record of the work carried out by me during my paid internship at **Saundarya App, Mumbai**.

The work presented in this report is based on my learning, implementation, and contribution during the internship period from **1 May 2026 to 30 June 2026**. This report has been prepared for academic submission and has not been submitted elsewhere for any other degree or diploma.

Place: RGUKT

Date: 14 July 2026

Shaik Hasmi roohy
ID No: R210448

CERTIFICATE

This is to certify that **Shaik Hasmi roohy**, ID No. **R210448**, from the Department of Computer Science and Engineering, Rajiv Gandhi University of Knowledge and Technologies, has successfully completed the paid summer internship project titled **"AI-Powered Saloon App"** at **Saundarya App, Mumbai**.

The work was carried out under the guidance of **Mr. A. Mahendra**, Project Internal Guide, Department of Computer Science and Engineering. The report is approved for submission as part of the internship evaluation.

Signature of Internal Guide	Signature of HOD	Signature of External Examiner
Mr. A. Mahendra	Dr. Ch. Ratna Kumari	Examiner
Assistant Professor	Assistant Professor	
Department of CSE	Department of CSE	

INTERNSHIP COMPLETION CERTIFICATE

Certificate issued by Saundarya App, Mumbai, for successful completion of the internship.

ACKNOWLEDGEMENT AND CONTENTS

I express my sincere gratitude to Rajiv Gandhi University of Knowledge and Technologies and the Department of Computer Science and Engineering for providing the opportunity to complete this internship. I am thankful to my project internal guide **Mr. A. Mahendra** for his guidance and support.

I also thank **Saundarya App, Mumbai**, and **Mr. Kamlesh Gupta**, Founder and CEO, for giving me an opportunity to work as a paid Flutter App Development intern and contribute to AI-powered features in the Saundarya application.

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ABSTRACT

The AI-Powered Saloon App is a Flutter-based mobile application developed to provide an interactive virtual beauty and salon try-on experience. The application allows users to upload or capture a photo and preview styles such as hairstyles, beard styles, nail art, tattoos, and mehndi patterns. It aims to reduce uncertainty before choosing a salon service by giving users a visual preview on their own image.

During the internship, the main focus was on building a user-friendly AR mirror workspace, improving gallery and camera flows, implementing model selection for free 2D try-on and premium AI try-on, and creating a premium checkout process. The system uses Flutter for the mobile interface, Google ML Kit for detection support, and a NestJS backend structure for premium AI image generation.

The project provided practical exposure to mobile UI design, image processing, application state management, payment-flow prototyping, and AI feature integration. The final application demonstrates how beauty technology can combine quick local previews with premium AI workflows for more realistic results. The internship also improved my understanding of handling real device testing, UI feedback, image editing tools, and user-friendly navigation in a Flutter application.

Keywords

Flutter, AI Salon, Virtual Try-On, AR Mirror, Google ML Kit, Image Editing, Premium Checkout, Saundarya App.

INTRODUCTION AND ORGANIZATION PROFILE

Customers often want to preview salon services before applying them in real life. The AI-Powered Saloon App addresses this requirement by providing a mobile platform where users can preview styles on their own images. In the current digital environment, customers expect fast, visual, and personalized experiences. A virtual try-on application helps users compare multiple styles before selecting a final look.

Saundarya App, Mumbai is a beauty and salon-focused application initiative where the internship was completed in the domain of Flutter App Development. The organization focuses on AI-powered beauty experiences and mobile-first salon workflows. During the internship, I worked on improving the app interface, developing try-on flows, and preparing premium feature screens for a complete demo.

My role involved understanding requirements, converting them into Flutter screens, testing the app on Android, and improving the user experience based on feedback. The work included both free preview features and premium AI-based feature flows.

Role	Flutter App Development Intern
Organization	Saundarya App, Mumbai
Internship Type	Paid Internship
Duration	1 May 2026 to 30 June 2026
Project	AI-Powered Saloon App
Guide	Mr. A. Mahendra

PROJECT OBJECTIVES AND REQUIREMENTS

The main objective of the project is to provide users with a simple and interactive way to preview beauty and grooming styles before applying them in real life. The app is designed to support both local 2D try-on and premium AI-assisted image editing.

Objectives

- To build an AR mirror workspace for photo-based virtual try-on.
- To support gallery upload and camera capture for user photos.
- To implement 2D overlays for hairstyle, beard, nail art, tattoo, and mehndi categories.
- To provide manual adjustment options such as move, resize, rotate, opacity, and reset.
- To design premium feature selection and checkout flows.
- To add save, share, full-screen preview, and edit options for generated results.
- To keep the interface understandable for users who are not technically trained.

Functional Requirements

- User should be able to upload an image from gallery or capture a new image from camera.
- User should be able to choose models from hair, beard, nails, tattoo, and mehndi categories.
- User should be able to drag, resize, rotate, and adjust selected overlays.
- User should be able to open a result in full screen and access download, share, edit, and AI edit options.
- Premium screen should allow multiple feature selection and calculate total amount.
- Checkout screen should support UPI, cards, net banking, and PayPal options.

SYSTEM REQUIREMENTS

The system requirements were selected according to the needs of Flutter mobile development and Android testing. Since the application deals with images, camera, ML-based detection, and Android build tools, enough storage and memory are important for smooth development.

Hardware Requirements

Component	Requirement
Processor	Intel i3 or above
RAM	8 GB recommended
Storage	10 GB free space
Device	Android phone/emulator

Software Requirements

Software	Purpose
Flutter & Dart	Mobile app
Android SDK	Build/testing
Google ML Kit	Detection
Node.js/NestJS	Backend

Development Dependencies

- **image_picker:** used for gallery and camera image selection.
- **google_mlkit_face_detection:** used for identifying face position.
- **google_mlkit_face_mesh_detection:** used for improved facial placement support.
- **google_mlkit_pose_detection:** used for tattoo and mehndi body placement support.
- **gal and share_plus:** used for saving and sharing try-on results.
- **url_launcher:** used for payment intent flows such as UPI and PayPal.

SYSTEM DESIGN

The application follows a modular Flutter architecture. The user opens the AI Virtual Try-On workspace, uploads a photo, selects a category, chooses a model, and previews the result. The result can be edited, saved, shared, or opened in the premium AI workflow.

Workflow

Step	Description
1	User uploads a photo from gallery or captures through camera.
2	The app detects face, mesh, pose, or hand landmarks depending on category.
3	User selects hairstyle, beard, nails, tattoo, or mehndi model.
4	The selected overlay is placed on the image and can be adjusted manually.
5	User saves, shares, edits, or opens the current image in Premium AI.

Architecture Explanation

The application separates the main try-on screen, premium screens, detection services, and rendering logic. This makes it easier to maintain the app and add new categories later. The AR mirror workspace handles user interaction, while renderer and detection services support overlay placement.

- **Presentation Layer:** Flutter screens, model cards, editor UI, premium checkout UI.
- **Service Layer:** Face detection, advanced detection, and overlay rendering services.
- **Data Layer:** Local try-on model definitions and premium AI repository structure.
- **Backend Layer:** NestJS endpoint preparation for premium AI image generation.

IMPLEMENTATION DETAILS

The implementation was mainly done using Flutter and Dart. The app contains a main AI Virtual Try-On workspace where users can switch between 2D Try-On and AI Premium modes. In 2D mode, the app applies local model overlays on the user image. In AI Premium mode, the selected image can be prepared for premium AI editing.

Technologies Used

- **Flutter and Dart:** used for UI, navigation, state handling, and mobile interaction.
- **Google ML Kit:** used for face detection, face mesh, pose detection, and segmentation support.
- **Image Picker, Gal, Share Plus:** used for gallery, camera, save, and share operations.
- **URL Launcher:** used for UPI and PayPal payment-related flows.
- **NestJS Backend:** prepared for premium AI image generation endpoint structure.

Implementation Highlights

The gallery flow was improved so that selected media first opens in a preview screen. From there, the user can decide whether to use the image in the try-on panel, download it, share it, delete it, or open it in the AI premium workflow. This avoids directly placing every selected gallery image into the panel.

The edit screen includes tools such as crop, cut, color, draw, rotate, and mirror. The editor allows users to make basic visual changes before saving the image back to the main result. The premium section includes multi-select feature pricing and checkout with payment method accordion logic.

MODULES DEVELOPED

Modules Developed

- **Free AR Mirror:** Hairstyle, beard, nail, tattoo, and mehndi previews.
- **Gallery and Preview:** Full-screen preview with download, edit, share, AI edit, and delete actions.
- **Edit Image:** Crop, cut, color, draw, rotate, mirror, undo, redo, and save tools.
- **AI Assistant:** Style description, model variations, selfie comparison, tattoo inspiration, and AI edit route.
- **Premium Checkout:** Feature selection with UPI, Cards, Net Banking, and PayPal payment options.

Module Description

The Free AR Mirror module forms the core part of the project. It lets users preview different salon styles without needing backend processing. The model cards are organized by category so users can quickly switch between hair, beard, nails, tattoo, and mehndi.

The AI Assistant module was added to make the app feel more guided. It accepts a style description, suggests matching model variations, supports selfie comparison, and connects the current image to the premium AI edit path. This module improves the overall user experience because users can explore styles with less manual searching.

The Premium Checkout module provides a structured payment screen. Wallets were removed as per the latest requirement, and the remaining methods include UPI, Cards, Net Banking, and PayPal. The selected payment method expands while others remain collapsed.

TESTING, RESULTS AND LEARNING OUTCOMES

Testing Summary

Feature	Expected Result	Status
Photo upload/camera	Image appears in try-on panel	Completed
Model selection	Overlay appears and can be adjusted	Completed
Edit result	Editor opens with working tools	Completed
Save/share	Result can be saved or shared	Completed
Checkout	Selected payment method expands properly	Completed

Results

The project resulted in a functional Flutter application that demonstrates salon style preview features. Users can upload images, apply models, adjust overlays, edit results, save images, share outputs, and explore premium AI workflows. The app also includes a checkout screen suitable for demo and development presentation.

Learning Outcomes

- Gained practical experience in Flutter UI development and state handling.
- Learned how to work with image upload, camera capture, save, and share flows.
- Understood the importance of user-friendly placement and edit tools in visual apps.
- Learned how ML-based detection can help in face, pose, and hand-related features.
- Improved debugging skills while testing Android builds and resolving UI issues.

CONCLUSION, FUTURE SCOPE AND REFERENCES

Conclusion

The AI-Powered Saloon App successfully demonstrates an interactive salon try-on platform using Flutter, local overlay rendering, image editing, ML-based detection support, and premium AI workflow preparation. It helped in understanding how modern mobile applications can use AI-assisted features to improve user decision-making and engagement.

Future Scope

- Integrate production-ready AI image generation with stable backend credits and monitoring.
- Add real payment verification and secure subscription management.
- Improve natural blending for hairstyle, beard, tattoo, nail, and mehndi previews.
- Add salon booking, nearby salon suggestions, and appointment management.
- Support before/after comparison, social sharing, and creator marketplace features.

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